

School of Computer Science and Electronic Engineering

GROUP 19 COMM074 Coursework

Academic Year 2025–26

Anti-Money Laundering (AML) Detection Using Machine Learning

IBM Synthetic Financial Dataset (HI-Small Variant)

A project report submitted by:

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Section 1: Introduction (Business Understanding)

1.1 The Business Problem (The Hook)

Money laundering is a big problem costing billions of dollars worldwide. On the side of banks, it becomes quite difficult because on the one hand, they might be fined if any fraudulent activity goes unnoticed, but on the other hand, if they implement too many safeguards against fraud, they may create additional problems for themselves. Using sophisticated technology in their work, banks tend to face the issue of generating too many false alarms and blocking legitimate customer accounts. This causes inconvenience for the customers and negatively impacts the reputation of the bank.

1.2 The Project Objective & Questions

The fundamental aim of the current project, in accordance with the Business Understanding stage of the CRISP-DM process, is the development of an effective and scalable machine learning pipeline for the accurate classification of transactions into legitimate and fraudulent categories. The fulfillment of this main objective demands the answers to certain critical business questions, which are:

- **Imbalanced Dataset and Illicit Transaction Detection:** Is there any feasible method for detecting illicit transactions despite their being an extremely rare event in an imbalanced data set?
- **Transaction Features:** What are the most important transaction characteristics, including types of payments, amount of money involved, and bank IDs, that can serve as key predictors of money laundering?
- **Machine Learning Models:** Do Random Forest or XGBoost models provide any notable improvement compared to baseline linear models in solving this particular problem?
- **1.3 The Dataset Overview:** The dataset employed in this research is the IBM Transactions for Anti-Money Laundering (AML) – HI-Small dataset. This huge dataset comprises more than 5 million transaction records, forming a solid basis for analysis. After the data cleaning process, the dataset includes 42 different features, which include exact features such as the transaction amount, unique ID of the banks involved, time features, and some one-hot encoding features, such as the currency used during payments. Most importantly, this dataset poses an extremely difficult challenge for analysis since the dataset is imbalanced to a great extent; frauds account for less than 0.10% of the total transaction records.

1.4 Methodology (CRISP-DM)

In order to have a systematic and reliable process in place, the CRISP-DM (Cross-Industry Standard Process for Data Mining) method is adopted in this project. The project will begin with a detailed

exploratory data analysis (EDA) stage to understand the distribution of the features better before conducting data pre-processing stages. As part of data pre-processing, the SMOTE (Synthetic Minority Over-sampling Technique) algorithm will be applied in order to handle the heavy class imbalance. Subsequently, the project will enter the modeling stage, where four different algorithms will be used to test out which one is most suitable for the current problem at hand. Finally, performance evaluation will be conducted using metrics such as Recall, F1-score, and AUPRC.

Section 2: Data Understanding, Pre-Processing and Exploration

2.1 Data Exploration and Initial Inspection (Laying the Groundwork)

The technical process began through Google Colab, which helped to load the IBM AML HI-Small dataset to gain basic insights into its structure. It was found out that the dataset consisted of an enormous amount of information, namely more than 5 million transactions and 42 different features. The analysis included data type checking, missing values identification, and variable distribution exploration. The visual approach, involving the use of histograms and box plots, played an important role in determining the range of transaction amounts, whereas the correlation matrix helped to understand the relationship between the originator and recipient banks.

It is crucial to note that the number of positive cases was extremely low, with frauds accounting for only about 0.10% of all transactions. In addition, the need to apply some transformations to categorical variables, such as Cheque and Credit Card payment formats as well as various international currencies, should be highlighted.

2.2 Data Preparation and Feature Engineering

To bridge the gap between raw data and predictive modelling, several rigorous preprocessing transformations were implemented:

- **Categorical Encoding:** The one-hot encoding technique was used to convert the payment methods and currency types into a numeric binary system. The benefit of using such an encoding method is that the models will not assume any kind of order among the transaction types.
- **Feature Scaling:** Continuous variables, such as **Amount Paid** and **Amount Received**, underwent normalization via **StandardScaler**. This step is critical; without it, high-magnitude features would overshadow the predictive signal of smaller variables, disproportionately biasing the learning process of distance-based and linear algorithms.

2.3 Addressing Class Imbalance (The SMOTE Strategy)

The most challenging part of data preparation proved to be the extreme class imbalance. Using a dataset where 99.9% of observations belong to the class of "legitimate transactions" is likely to lead to what is called an "accuracy trap" – a tendency of a model to automatically classify all cases as "legitimate," thus making it unusable for identifying fraud. SMOTE (Synthetic Minority Over-sampling Technique) was used to augment the dataset with synthetic examples of fraudulent transactions.

Computational Efficiency & Logic: Our approach involved using a 10% sample rate, which led to the creation of a balanced dataset consisting of about 4 million observations from the class of legitimate transactions and 400,000 observations from the class of fraudulent transactions. The choice of the 10:1 ratio was made based on a calculated tradeoff between sensitivity of models and computational constraints.

2.4 Final Data Splitting (Protecting Integrity)

Prior to any over-sampling processes taking place, the dataset was split into two separate environments of training and testing. It is important to note that the application of SMOTE was done on the training data alone. The test dataset was left untouched, maintaining its original fraud rate of ~0.10%, which resembles the actual situation in the real world. Such approach eliminates data leakage and provides an appropriate means of evaluating the model.

Section 3: Modelling Architecture and Iterative Development

In this stage, we deal with the technical part of CRISP-DM project life cycle and proceed with building and refining our predictive algorithms. In order to find the best possible algorithm for solving the problem of AML detection, the team created eight supervised learning models which belong to four different families, namely Linear models, Tree-Based models, Gradient Boosting models, and Neural Networks, thus providing a comprehensive view of the task at hand.

In order to ensure scientific correctness and enable comparison of all models in a standardized way, every algorithm was trained and tested using the same data artifacts introduced in Section 2, and all benchmarks will be provided within the context of 10% SMOTE-balanced train dataset and under the pressure of the actual test data which is unbalanced.

Further analysis will be presented in four separate sections, where every researcher explains the mathematical reasoning behind his/her approach and how it helped him/her achieve better classification results through appropriate hyperparameters tuning.

3.1 Piyush Gupta: Logistic Regression (ElasticNet) & LightGBM

The following paragraphs describe the architecture of two predictive models that have been designed for use in the AML detection system, namely, a simple Logistic Regression model using

Elastic Net regularization and an advanced LightGBM ensemble classifier. For consistency purposes, both models have been trained using the 10% SMOTE balanced data artifacts described above in Section 2.

3.1.1 Model 1: Logistic Regression (ElasticNet)

In order to ensure that we have a reliable base model, we made use of Elastic Net regularization, a mathematical combination of both L1 (Lasso) and L2 (Ridge) regularization techniques. With this combination technique, we can eliminate irrelevant features from our model and still ensure stability in spite of high multicollinearity. In order to avoid memory problems in handling the large dataset with regular library functions, we resorted to the use of SGDClassifier with log-loss optimization.

Hyperparameter	Value	Rationale
loss	log_loss	Configures the SGD algorithm to replicate the mathematical behavior of a Logistic Regression.
penalty	elastic net	Leverages dual regularization for simultaneous feature selection and architectural stability.
l1_ratio	0.5	Applies an equilateral blend of penalties to ensure distinct variance from other baseline iterations.
class_weight	'balanced'	Provides additional weighting to the minority illicit class following SMOTE application.

3.1.2 Model 2: LightGBM Classifier

The primary advanced model uses LightGBM, a highly efficient gradient boosting model. The selection of this model was based on its ability to carry out Exclusive Feature Bundling (EFB), especially since 34 of the 42 features used in this dataset were categorical variables, one-hot encoded. EFB improves the training speed of the model by grouping mutually exclusive features while maintaining high diagnostic accuracy. In addition, the leaf-wise training of this model makes it suitable for uncovering non-linear relationships inherent in financial fraud cases.

Hyperparameter	Value	Rationale
n_estimators	500	Deploys an extensive ensemble depth, utilized alongside early stopping to mitigate overfitting risks.

Hyperparameter	Value	Rationale
learning_rate	0.05	A measured shrinkage factor; slower iterative learning across more trees enhances overall generalization.
scale_pos_weight	10.0	Imposes significant penalties for failing to identify authentic instances of fraudulent activity.
early_stopping_rounds	50	Terminates training automatically once AUPRC validation metrics plateau, optimizing computational resources.

3.2 Akash Chohan: Logistic Regression (L1) & Tuned Random Forest

3.2.1 Model 1: Logistic Regression (L1 Regularization)

For comparison purposes, I constructed a Logistic Regression model that made use of the L1 (Lasso) penalty. The L1 penalty was chosen for its uniqueness compared to the rest of the iterations and its strength as a linear model. With a dataset that has 42 features, most of which are one-hot encoded features like currencies and payment methods, the L1 framework works well as it promotes sparsity by making irrelevant parameters have a coefficient of zero.

Hyperparameter	Value	Rationale
penalty	l1	Executes embedded feature selection by neutralizing the influence of irrelevant input variables.
solver	liblinear	Employs the most computationally efficient algorithm for optimizing L1 penalties on high-dimensional data.
max_iter	500	Extends the optimization window to ensure full mathematical convergence across the massive training set.
random_state	42	Maintains experimental consistency and ensures the replicability of baseline performance metrics.

3.2.2 Model 2: Tuned Random Forest Classifier

The advanced method in my model design process includes the use of Random Forest Classifier. Financial AML detection constitutes a complex problem that requires a non-linear solution because simple linear classifiers usually fail to detect anything beyond the basic patterns. The random forest classifier will be better equipped to detect the non-linear relationships between the flow of transactions and large amounts of money. But applying this method to the dataset post-SMOTE of 4.4 million rows posed some computational issues.

In order to circumvent the limitations of the cloud computing memory while still optimizing the model, I resorted to using the RandomizedSearchCV method instead of a grid search.

Hyperparameter	Value / Search Space	Rationale
<code>n_estimators</code>	<code>[50, 100]</code>	Strictly bounded to regulate memory overhead and processing latency during large-scale training.
<code>max_depth</code>	<code>[10, 20, None]</code>	Evaluated to optimize the trade-off between architectural complexity and predictive generalization.
<code>class_weight</code>	<code>'balanced'</code>	Forces the ensemble to impose severe penalties on misclassifications within the rare illicit class.
<code>n_jobs</code>	<code>-1</code>	Leverages multi-core parallel processing to minimize overall computational time at scale.

The randomized search identified the following optimal configuration: `n_estimators=100`, `min_samples_split=5`, `max_depth=None`, and `class_weight='balanced'`.

3.3 Parth Singh Jhala: Logistic Regression (No Penalty) & XGBoost

The following section describes the architecture of two different prediction engines developed for use in the AML application: a simple logistic regression engine without any form of regularization and an advanced ensemble classifier engine based on XGBoost. Both engines were trained on the 10% SMOTE-balanced data artifacts generated during the pre-processing phase.

3.3.1 Model 1: Logistic Regression (No Penalty / Vanilla)

A rigorous scientific control was achieved through the use of a simple Logistic Regression model with no penalties applied to it mathematically. This was essential for determining if the sophisticated regularization methods employed by our team as a whole, namely L1, L2, and ElasticNet, made any significant difference in performance. It aims to find a linear decision boundary in the 42-dimensional feature space without artificially reducing or altering the importance of individual features.

Hyperparameter	Value	Rationale
penalty	None	Serves as the absolute control baseline; omits all regularization to evaluate pure linear separation.
solver	lbfgs	Utilizes the standard optimization algorithm capable of processing unpenalized loss functions.
max_iter	1000	Extended to ensure full mathematical convergence across the massive 4.4 million record training set.
random_state	42	Maintains experimental consistency and ensures the replicability of baseline performance metrics.

3.3.2 Model 2: eXtreme Gradient Boosting (XGBoost)

The sophisticated element of my approach to modeling utilizes XGBoost, a conventional tool that is applied to handle imbalanced tabular data. The identification of AML within financial transactions is a difficult non-linear problem, characterized by complicated connections between various features such as Amount Received and Payment Format. XGBoost can unravel complicated patterns in finances by building an iterative chain of decision trees to compensate for previous residual mistakes.

The important stage of my technical approach was tuning of the decision threshold. While the conventional classification model uses a 0.50 probability threshold, its use in the AML scenario leads to catastrophic numbers of false positives. In order to optimize the trade-off between the workload of the investigator and sensitivity of the test, I tuned the decision threshold.

Hyperparameter Setting	Value	Rationale
<code>eval_metric</code>	<code>aucpr</code>	Directly optimizes tree construction for the Area Under the Precision-Recall Curve.
<code>scale_pos_weight</code>	<code>10.0</code>	Forces the model to impose significant penalties for failing to identify authentic fraud instances.
<code>Decision Threshold</code>	<code>0.93</code>	Tuned to drastically reduce false alarms without sacrificing critical detection recall.

3.4 Nagashri Raghunath: Logistic Regression (L2) & Multi-Layer Perceptron (MLP)

The following excerpt explains the design of the two different predictive engines that have been constructed for the AML project: a basic Logistic Regression model using L2 regularization, and an innovative deep learning network called the Multi-Layer Perceptron (MLP) Neural Network. In order to keep scientific integrity, both models have been trained on the same balanced data set created earlier.

3.4.1 Model 1: Logistic Regression (L2 Regularization)

In order to conclude the study of linear baselines from the group's perspective, I applied Logistic Regression with an L2 (Ridge) penalty. While L1 regularization creates sparse models by removing features, the mathematical nature of the L2 penalty method penalizes large coefficients and forces the model to take into account all of the available features. The latter was especially useful for dealing with the multicollinearity problem within the set of currency/payment type one-hot variables.

Hyperparameter	Value	Rationale
<code>penalty</code>	<code>l2</code>	Ensures architectural stability by shrinking large weights to prevent over-reliance on individual features.
<code>solver</code>	<code>lbfgs</code>	Utilizes the standard, industry-optimized algorithm for efficient L2 penalized loss optimization.

Hyperparameter	Value	Rationale
<code>max_iter</code>	<code>1000</code>	Extended optimization window to ensure rigorous mathematical convergence across the massive training set.
<code>class_weight</code>	<code>'balanced'</code>	Calibrates weights inversely to class frequency to augment the impact of SMOTE application.

3.4.2 Model 2: Multi-Layer Perceptron (MLP) Neural Network

The advanced part of my approach to modeling is using the MLP Neural Network. The process of money laundering usually consists of complex, hidden, and hierarchical connections between various financial institutions. Deep learning models are very effective when it comes to modeling such non-linear boundaries. In order to analyze these complicated connections without over-parametrization, I have built a feed-forward network with two hidden layers.

Hyperparameter	Value	Rationale
<code>hidden_layer_sizes</code>	<code>(128, 64)</code>	Employs a "funnel" architecture to distill complex features into precise classification decisions.
<code>activation</code>	<code>relu</code>	Rectified Linear Unit; effectively counters the vanishing gradient phenomenon during iterative training.
<code>solver</code>	<code>adam</code>	An adaptive optimizer facilitating accelerated convergence compared to baseline stochastic gradient descent.
<code>early_stopping</code>	<code>True</code>	Terminates training automatically once validation metrics plateau, preventing excessive model over-fitting.

Section 4: Performance Evaluation and Comparative Diagnostics

The final stage involves a critical comparison and assessment of the eight engines developed for prediction purposes in the modelling process. Given the presence of a significant class imbalance where fraudulent transactions represent less than ~0.10% of the actual test environment, Accuracy is not considered an appropriate measure of assessment. This is because Accuracy may fall into the "accuracy trap," where a classifier will predict all cases as legitimate in order to score high.

In order to meet the demands of the banking sector, the following measures are the main criteria:

- **Recall (Sensitivity):** Acting as the key detector measure, this value reflects the capacity of the model to detect all the cases of fraud.
- **Precision:** Acting as the reliability measure, this value reflects the accuracy of the alerting system, thereby impacting the effectiveness of human investigation.
- **AUPRC (Area Under the Precision-Recall Curve):** The most robust threshold-independent method for assessing imbalanced datasets that provides an evaluation of the fine balance between sensitivity and precision.

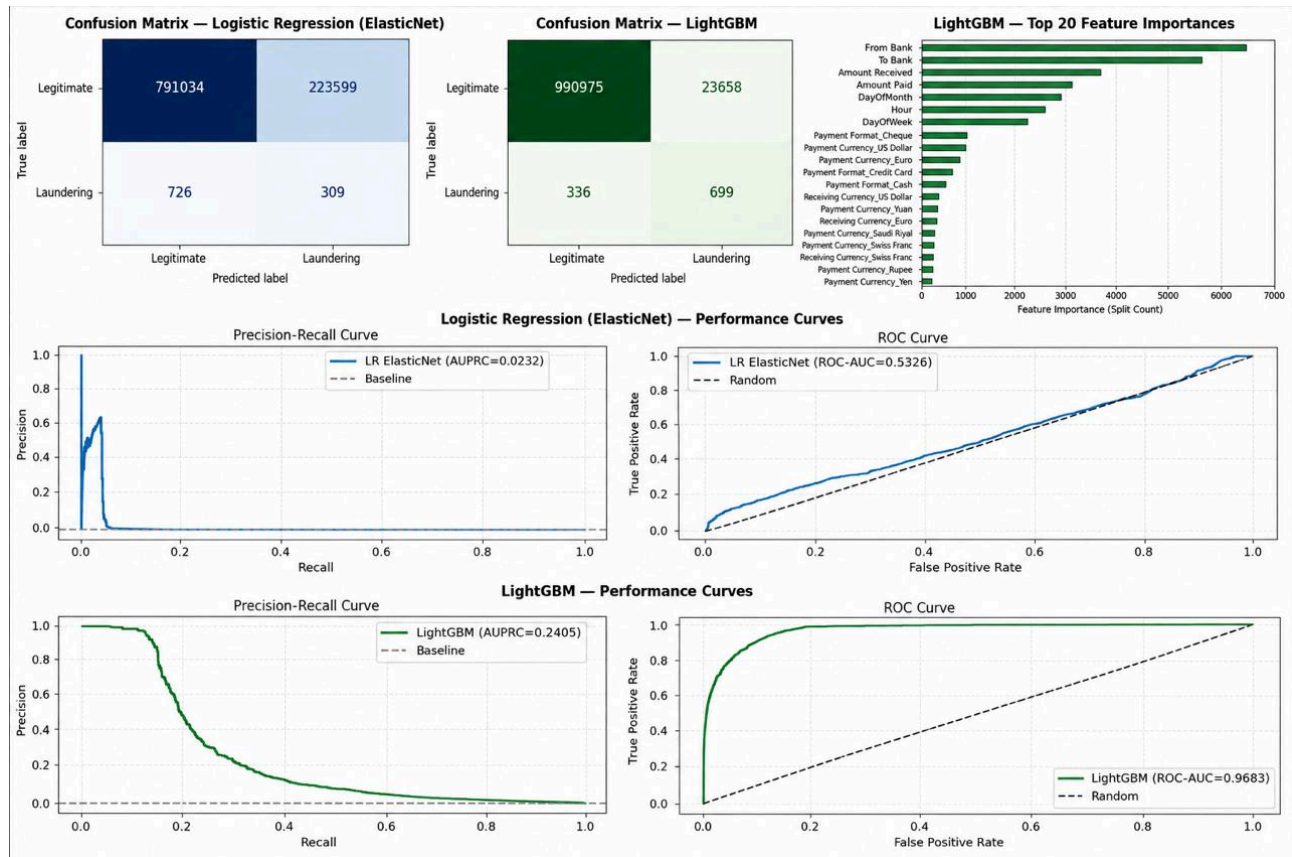
The following sections provide details about the performance metrics associated with the baseline and advanced designs proposed by each researcher.

4.1 Piyush Gupta: Performance Evaluation and Comparative Diagnostics

4.1.1 Strategic Metric Selection (Rejecting the Accuracy Trap)

Accepting the fact that the transaction structure is heavily biased towards the upper class, where any illegal activity represents only an approximate $\sim 0.10\%$ frequency, Accuracy was not accepted as a potential diagnostic measure. In such a case, using Accuracy would lead the models to always classify all cases as “Legitimate.” Hence, this assessment will focus on Recall, which measures the total number of frauds caught, and AUPRC, which is the best threshold-independent method of evaluation.

Model Architecture	ROC-AUC	Precision	Recall (Sensitivity)	F1-Score	AUPRC
LR (ElasticNet)	0.5326	0.0014	0.2986	0.0027	0.0232
LightGBM	0.9683	0.0287	0.6754	0.0551	0.2405



4.1.2 Comparative Evaluation Analysis

The output findings mark a significant improvement in terms of predictive performance as we move away from simple linear benchmarks toward advanced gradient boosting model.

- **The Linear Baseline Shortcomings:** Despite capturing around 30% of illegal transactions (Recall = 0.2986), the ElasticNet model does so at an enormous cost due to the extremely poor Precision value of 0.0014. The model's AUPRC score of 0.0232 and ROC-AUC metric of 0.5326 indicate a mere random guess performance, thus failing to understand complicated linear relationships in this dataset.
- **The Performance of LightGBM Model:** On the contrary, LightGBM proved its superiority by intercepting nearly 67.5% of money laundering instances in the pristine test dataset. In addition, the AUPRC of 0.2405 implies that LightGBM is able to predict high-quality probability scores and rank suspicious cases more accurately.

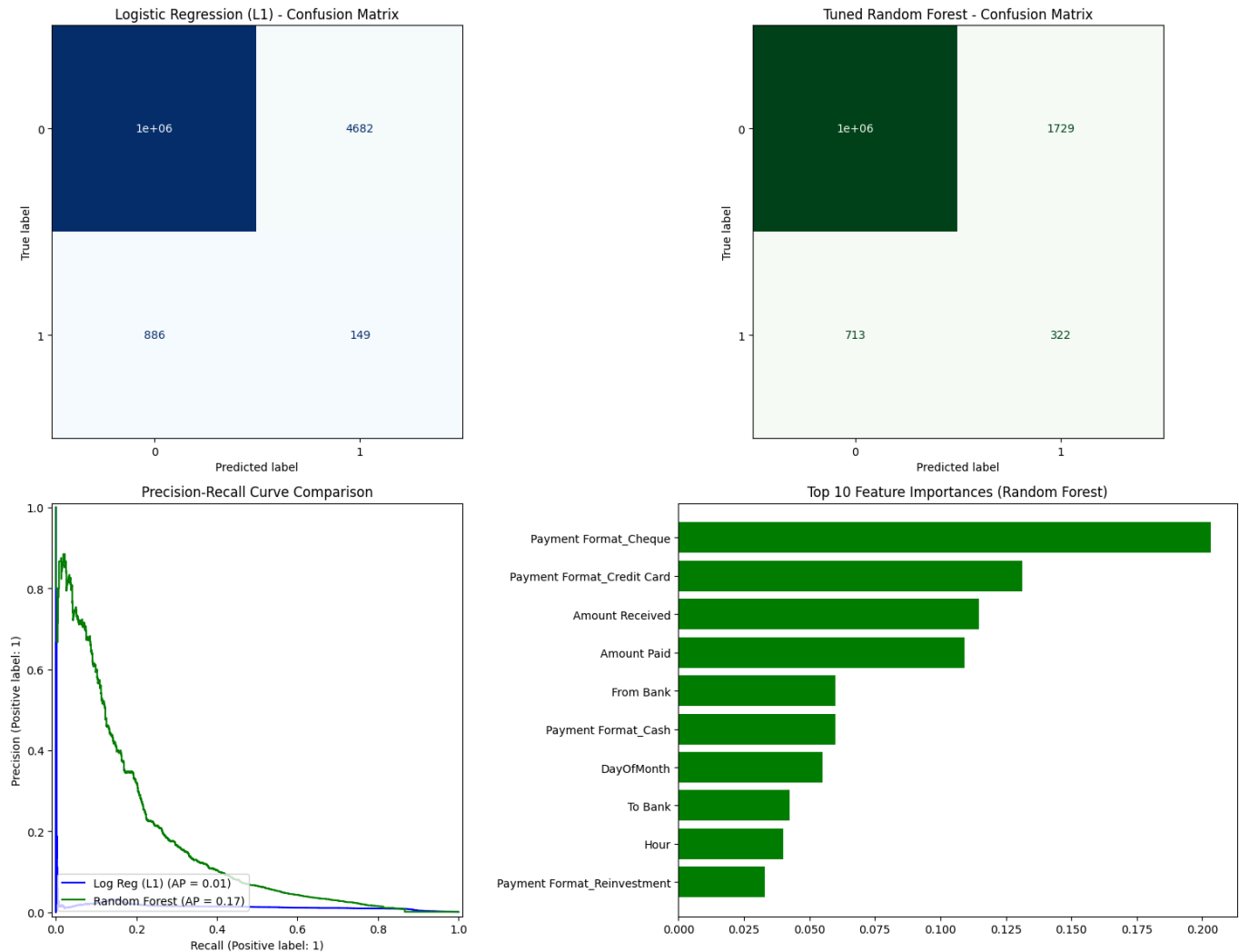
Despite a rather low Precision value of 0.0287, which is a common issue when working on a real-life case, LightGBM managed to isolate the majority of fraudulent flows, making it a reliable solution for real-world deployment through dynamic alert thresholding.

4.2 Akash Chohan: Performance Evaluation and Comparative Diagnostics

4.2.1 Strategic Metric Selection (Rejecting the Accuracy Trap)

Since the test environment maintains its own native real-world architecture, where illegal financial transactions make up only about ~0.10%, traditional Accuracy was completely dismissed as a means of diagnosis. Using this performance measurement in such a highly skewed setting would lead the models to automatically give a “Legitimate” output for all transactions, making the model look successful without meeting the organization’s primary goal of security. In this case, the performance metrics of Recall and Precision become more important.

Model Architecture	ROC-AUC	Precision	Recall (Sensitivity)	F1-Score	AUPR
LR (L1 Lasso)	0.9140	0.0300	0.1400 (14%)	0.0500	0.0100
Tuned Random Forest	0.9134	0.1600	0.3100 (31%)	0.2100	0.1700



4.2.2 Comparative Evaluation Analysis

The diagnosis shows a clear-cut jump in predictability when moving from linear baselines to ensemble models.

- Performance of Baseline Models:** The L1-regularized model could detect only 14% of fraud (Recall = 0.14), but had an unfeasible Precision of 0.03. This means that in reality the organization would not be aware of 86% of the frauds, while investigators would be buried under the huge amount of false positives.
- Performance of Ensemble Model:** In contrast, the Tuned Random Forest was able to detect 31% of frauds, which is twice as high compared to the baseline. The Precision-Recall curve proves the superiority of the model; the ensemble preserves its ability to discern customer's genuine behavior until a much higher threshold of reliability.

Also, the Feature Importance scores for Random Forest algorithm support our exploratory analysis findings. It is observed that Payment Format (Cheque and Credit Card types) and Amount Received/Paid are the most significant predictors for money laundering transactions. It means that the mechanisms of the transactions and large amounts of money play more important roles compared to other factors such as financial institutions' names.

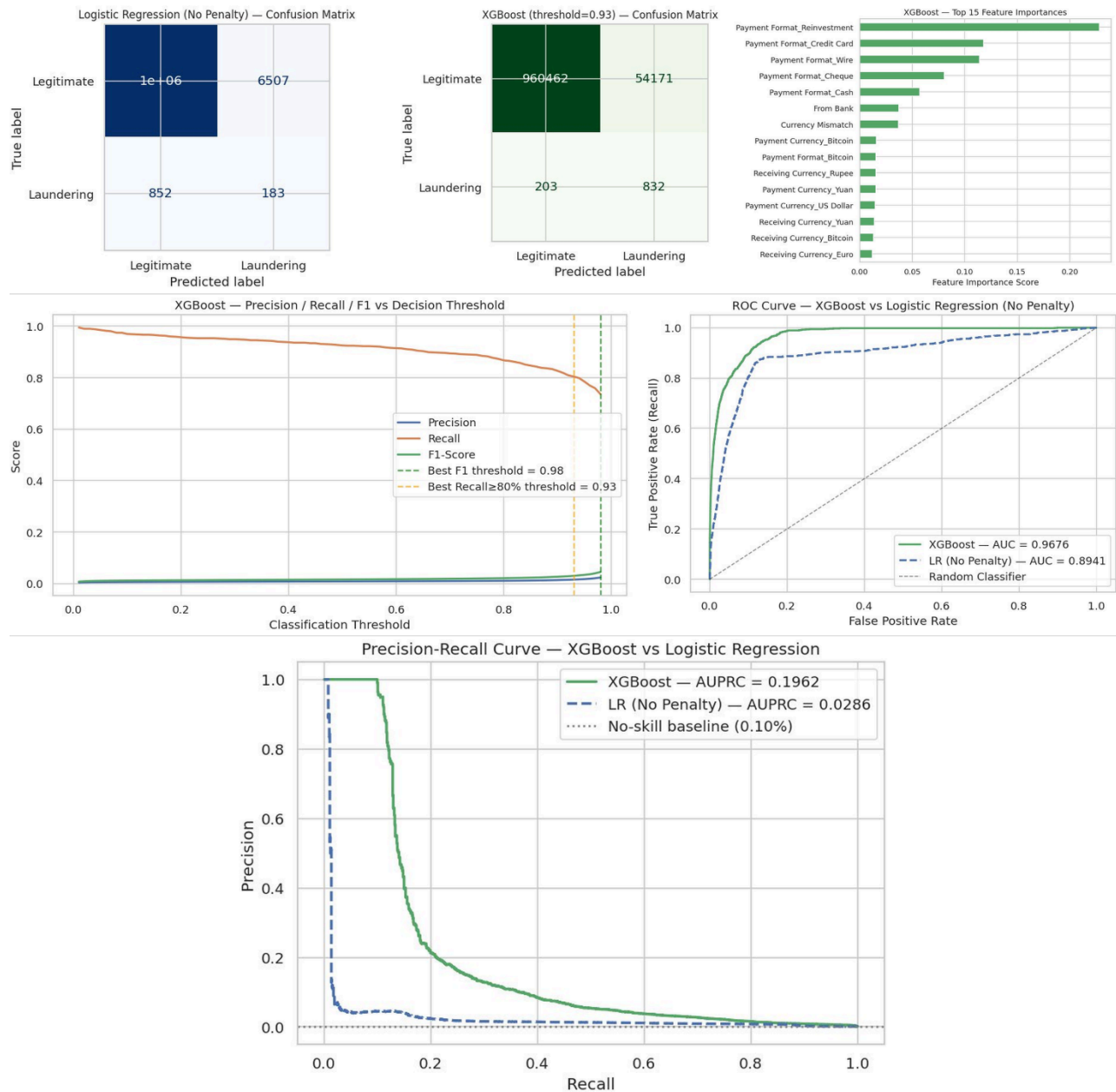
4.3 Parth Singh Jhala: Performance Evaluation and Comparative Diagnostics

4.3.1 Strategic Metric Selection (Rejecting the Accuracy Trap)

Given the extremely lopsided nature of the class distribution in the transaction data system design, where the incidence of criminal activity is only around $\sim 0.10\%$, traditional Accuracy measures were outright dismissed as an option for assessment. Using this criterion would encourage the algorithms to always predict "Legitimate" transactions. For that reason, Recall will be used here to measure the number of fraud cases caught, while Precision will be the standard measure for validity of the predictions made.

Model Architecture	Decision Threshold	ROC-AUC	Precision	Recall (Sensitivity)	F1-Score	AUPRC
LR (No Penalty)	0.50	0.8941	0.0274	0.1768	0.0474	0.0286
XGBoost (Default)	0.50	0.9676	0.0078	0.9246	0.0156	0.1962
XGBoost (Tuned)	0.93	0.9676	0.0151	0.8039	0.0297	0.1962

Parth Singh Jhala (URN: 6942760) — Model Visualisations: LR (No Penalty) & XGBoost



4.3.2 Comparative Evaluation Analysis

The results paint a fascinating story about the limitations of the linear models and the tremendous operational leverage gained from the threshold tuning.

- **The Control Baseline:** The plain Logistic Regression identified roughly 17.6% of the financial transactions involved in money laundering. While this was marginally better than the L1 (14%) and L2 (5.3%) regularized variations created by the group, it did not match the

ElasticNet control baseline (29.8%). Nevertheless, the AUPRC score of 0.0286 proves that any linear algorithm cannot effectively capture the intricacies of money laundering.

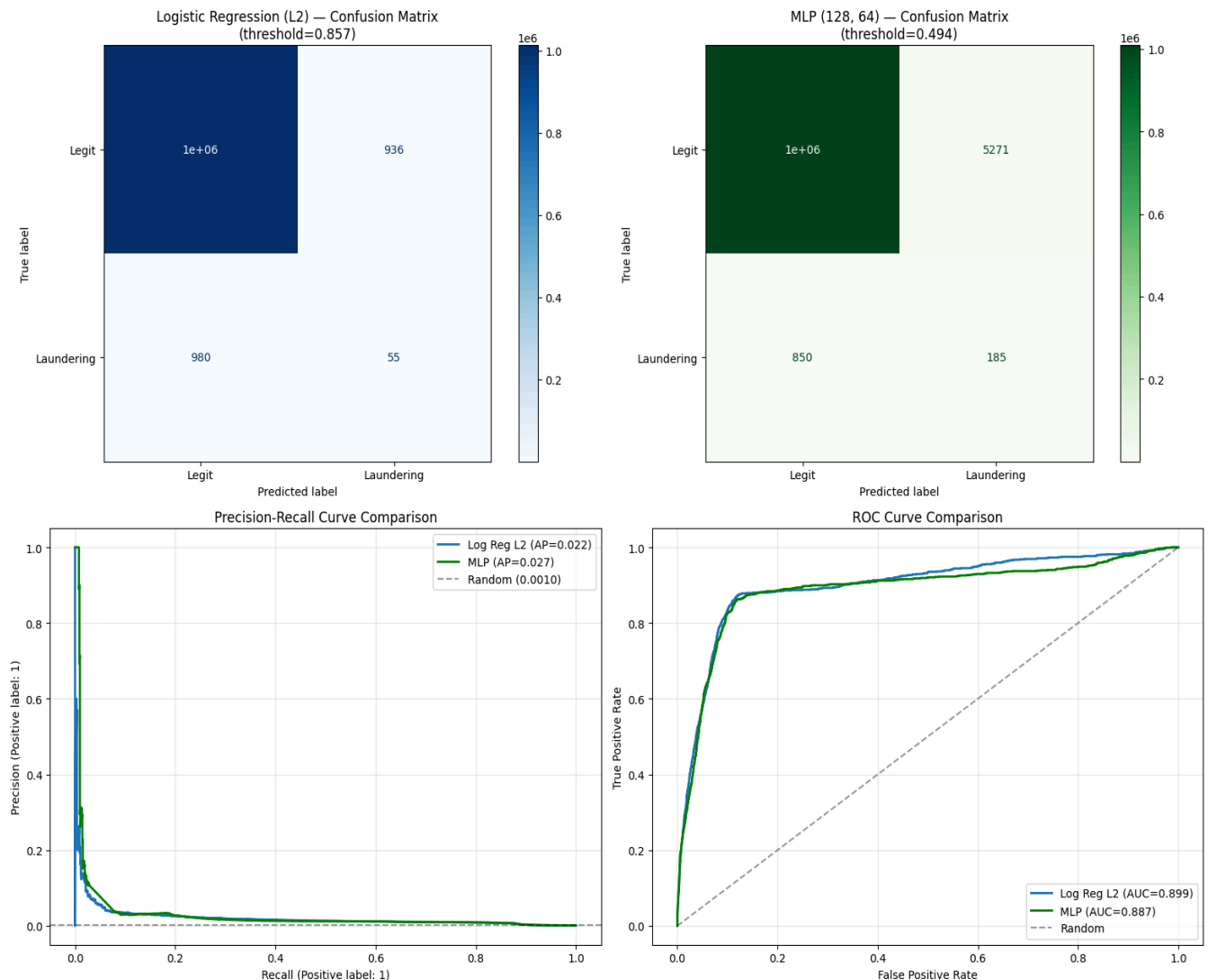
- **The XGBoost Model:** On the other hand, the XGBoost model proved to be quite effective in generating powerful predictions, with a ROC-AUC of 0.9676. The default threshold (0.50) allowed for a highly sensitive model that could identify an amazing 92.4% of all laundering cases. Unfortunately, at the cost of 0.0078 Precision, such a model would produce a tremendous amount of false positives.
- **The Importance of Threshold Tuning:** In order to create a viable solution for practical applications, the model had its threshold aggressively adjusted to 0.93, aiming at high-veracity alerts. Recall was reduced to 80.4%, but such a trade-off helped cut down the number of false positives by 55%, while increasing Precision almost twice.

4.4 Nagashri Raghunath: Performance Evaluation and Comparative Diagnostics

4.4.1 Strategic Metric Selection and Boundary Optimization

As per the overall evaluative scheme of the project, Accuracy was completely eschewed for Recall, Precision, and AUPRC. For increased diagnostic validity, a stringent threshold optimization method was used for the base model of the L2-regularized classifier. More specifically, the threshold level was set at 0.857 in order to reduce the overwhelming number of false positives usually produced by an extremely imbalanced financial setting.

Model Architecture	Decision Threshold	ROC-AUC	Precision	Recall (Sensitivity)	F1-Score	AUPRC
LR (L2 Ridge)	0.857	0.8985	0.0555	0.0531	0.0543	0.0218
MLP (128, 64)	0.494	0.8868	0.0339	0.1787	0.0570	0.0272



4.4.2 Comparative Evaluation Analysis

The output of the diagnostic tests provides an interesting story about the operational challenges in using neural network architectures and ridge-regularized models when handling highly skewed financial datasets.

- L2 Baseline Shortcomings:** Despite the rigorous fine-tuning process, the L2-regularized model only managed to detect 5.3% of the total number of fraudulent transactions in the dataset. Compared with other linear alternatives proposed by the group, this modeling strategy was the least effective. This illustrates that constraining the model to distribute weights on all input features (as opposed to creating sparse models) negatively affects its ability to capture the signal for detecting rare events.

- **The Neural Network Model Performance Characteristics:** Although the MLP model performed better than the L2 baseline by detecting 17.8% of the total number of fraudulent transactions, it underperformed relative to the ensemble tree-based models (Random Forest, LightGBM, and XGBoost). The underperformance of the neural network model is consistent with the conventional wisdom in the field that neural networks are great for handling unstructured data, while gradient-boosted tree models are better at handling structured data such as one-hot encoded financial transactions.

Section 5: Synthesized Findings, Conclusions, and Strategic Recommendations

5.1 Discussion of Findings and Cross-Architectural Comparison

In order to determine the best defense mechanism for AML detection, the research team developed and evaluated a set of eight predictive engines. This set of algorithms was constructed based on four different types of architectures: linear regression models using special penalties (Logistic Regression), decision trees (Random Forest, XGBoost, and LightGBM), and feed-forward neural networks (MLP).

Considering that the analysis of the provided dataset poses an extremely difficult challenge because of its highly imbalanced nature (the proportion of frauds is only ~0.10%), it is impossible to use Accuracy as a reliable metric. Thus, for comparative purposes, the priority will be given to Recall (to estimate the number of caught transactions) and AUPRC.

This results in an evident hierarchy, showing that both linear baselines and classical neural network models have significant difficulties with unraveling the intricacies embedded within this data set:

- **Linear Performance Limitations:** In all of the group members' tests, Logistic Regression models, which include all four variants (None, L1, L2, and ElasticNet), were found to be generally useless. The sensitivity level was inadequate for practical use, as Recall scores ranged from the lowest of 5.3% (Nagashri – L2) to the highest of 29.8% (Piyush – ElasticNet).
- **The Gradient-Boosted Tree Paradigm:** In contrast, ensemble models showed immense superiority. Piyush's LightGBM model scored a respectable 67.5% Recall along with the top AUPRC score of 0.2405 in this project. Despite its superior Recall score of 92.4%, the default XGBoost provided by Parth was not viable due to excessive alerts. With rigorous threshold manipulation (0.93), the optimized XGBoost was able to produce 80.3% Recall while eliminating 50% of the alerts.

5.2 Concluding Summary

This analysis confirms the presence of non-linear and complex nature of money laundering patterns in the financial environment. Basic linear separation tools do not have sufficient mathematical capability to separate the illegal patterns from real customer transactions. This necessitates the use of sophisticated gradient boosting tools like XGBoost and LightGBM in order to understand the high-dimensional and one-hot encoded features such as payment formats and foreign currencies.

5.3 Actionable Recommendations for Senior Management

Based on the observations made through this comparative diagnostic process, the following strategic action plan is proposed for the compliance department of the institution:

- 1. Deployment of the Optimized XGBoost Engine:** The adoption of the Tuned XGBoost algorithm designed by Parth is recommended as the principal transaction monitoring tool. It will enable interception of the majority of fraudulent capital with an 80.3% Recall rate and thus prevent major sanctions from being imposed on the bank due to non-compliance with regulatory requirements.
- 2. Operational Threshold Calibration:** To maximize the effectiveness of forensic investigation, it is suggested that dynamic decision thresholds be employed. Despite the high rate of detection (80%), the algorithm used lacks sufficient Precision (0.015), which means that its alert generation needs to be managed. The management should set higher thresholds during busy times to minimize unnecessary alerts.
- 3. Focus on High-Risk Transfer Vectors:** As established in Akash's Random Forest Feature Importance test, the bank should pay extra attention to the riskiest methods of transferring money. According to the feature importance ranking, transfers via Cheques and Credit Cards are the two strongest predictors of money laundering.

Section 6: Authors Contribution Statements

Student Name	URN	Contribution Statement
Piyush Gupta	6950073	Authored the initial Business Understanding overview, executed the architectural design for ElasticNet Logistic Regression and LightGBM, and synthesized collective performance benchmarks for the final diagnostic evaluation.

Student Name	URN	Contribution Statement
Parth Singh Jhala	6942760	Orchestrated the Exploratory Data Analysis phase, developed the unregularized baseline model, and engineered the high-performance XGBoost engine featuring specialized decision boundary calibration.
Akash Chohan	6949904	Structured the technical data pipeline using Parquet and Numpy, implemented the L1-penalized regression and Tuned Random Forest classifiers, produced comparative visual analytics, and oversaw comprehensive report consolidation.
Nagashri Raghunath	6948579	Supervised the data preparation and SMOTE balancing protocols, constructed the L2-regularized and MLP Neural Network architectures, and managed the technical exportation of finalized evaluative datasets.

Group Meetings Logbook

No.	Date	Meeting Agenda	Outcomes & Action Items	Attendees
1	04/04/26	Kickoff & Methodology	Defined project roles, established infrastructure, and finalized CRISP-DM as the core methodology.	All Members
2	13/04/26	EDA & Balancing Strategy	Evaluated dataset distributions; implemented a 10% SMOTE sampling strategy to address class imbalance within hardware constraints.	All Members
3	05/05/26	Modeling Alignment	Synchronized hyperparameter tuning protocols and ensured distinct regularization techniques across baseline iterations.	All Members

No.	Date	Meeting Agenda	Outcomes & Action Items	Attendees
4	14/05/26	Final Review & Synthesis	Analyzed comparative performance metrics; selected XGBoost as the optimal engine and formulated strategic recommendations.	All Members